

ASSEMBLE AI

AI-Powered Rent Collections & Leasing for Africa

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Confidential

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The Problem

South Africa has 4.5 million renting households generating over R480 billion in annual rental flows. At any given time, approximately 390,000 of these households are in arrears, representing R2.7 billion in outstanding unpaid rent.

The current collections process is manual, slow, and expensive. Property managers chase arrears through phone calls, letters, and attorneys. The average time from first missed payment to resolution is 6 to 18 months. Eviction through South African courts under the PIE Act is heavily regulated and can take over a year.

Meanwhile, South Africa's infrastructure challenges compound the problem. Load shedding disrupts call centres and payment systems. Manual EFT payments have failure rates of 30–40%. Property managers lose thousands of rands per unit in the gap between a tenant stopping payment and eventual resolution.

Key Market Data

Tenants in arrears nationally: 17% (~390,000 households)

Outstanding unpaid rent: R2.7 billion+

Average resolution time: 6–18 months

Manual EFT payment failure rate: 30–40%

AI-powered collections tools in SA: Zero

In the United States, EliseAI has built a \$2.2 billion company solving this exact problem with AI-powered collections and leasing. They reduce delinquencies by 52% per quarter and operate across 10% of all US apartments. Their collections product is their fastest-growing line.

No equivalent product exists in South Africa or anywhere on the African continent. The tools, the market, and the timing are aligned for Assemble AI to own this space.

The Product

Assemble AI is an AI-powered collections and leasing platform purpose-built for the South African rental market. It uses conversational AI across WhatsApp, voice calls, SMS, and email to automate the entire tenant communication lifecycle — from first rental enquiry to rent collection to arrears recovery.

The platform is built on three product layers, launched sequentially to maximise focus and speed to market.

Product 1: Collections AI (Launch Product)

An AI agent that contacts tenants in arrears via WhatsApp (primary channel), voice calls (escalation), and SMS/email (fallback). The agent operates 24/7, communicates in English, Zulu, Sotho, and Afrikaans, and handles the full collections workflow:

- Sends structured, POPIA-compliant payment reminders on an escalating cadence
- Negotiates payment plans conversationally, adapting tone based on payment history and arrears depth
- Sends secure payment links via WhatsApp for immediate settlement
- Integrates with DebiCheck for authenticated debit orders (90–95% collection success vs 60–70% for manual EFT)
- Tracks every interaction for legal compliance including letter of demand prerequisites and PIE Act timelines
- Generates legal-ready documentation when escalation to attorneys is required
- Provides a real-time dashboard showing who owes what, contact history, payment commitments, and recovery rates

Why WhatsApp First:

WhatsApp has 95%+ penetration in South Africa. It is the de facto communication channel for property managers and tenants alike. Unlike phone calls, WhatsApp is resilient to load shedding (async delivery), creates a persistent audit trail for POPIA compliance, and allows tenants to respond at their convenience. Voice calls serve as escalation for non-responders.

Product 2: Leasing AI (Month 6+)

An AI agent that responds to every rental enquiry from Property24, Private Property, and direct channels within 60 seconds via WhatsApp. Speed of first response is the single biggest predictor of leasing conversion, yet most South African agents respond in hours or days.

- Instant lead response across all channels (WhatsApp, SMS, email, webchat)
- Pre-screens tenants with automated income verification and TPN credit check integration
- Schedules property viewings and sends automated confirmations and reminders
- Processes rental applications digitally, end to end
- Generates POPIA-compliant lease agreements
- Manages move-in inspection workflows and deposit collection
- Supports 50+ written languages for international tenants

Expected Impact: Based on EliseAI benchmarks: 125% more prospects converted to viewings, 30% increase in lead-to-lease conversion rate, 15-day acceleration in renewal notices.

Product 3: Property Operations Platform (Month 12+)

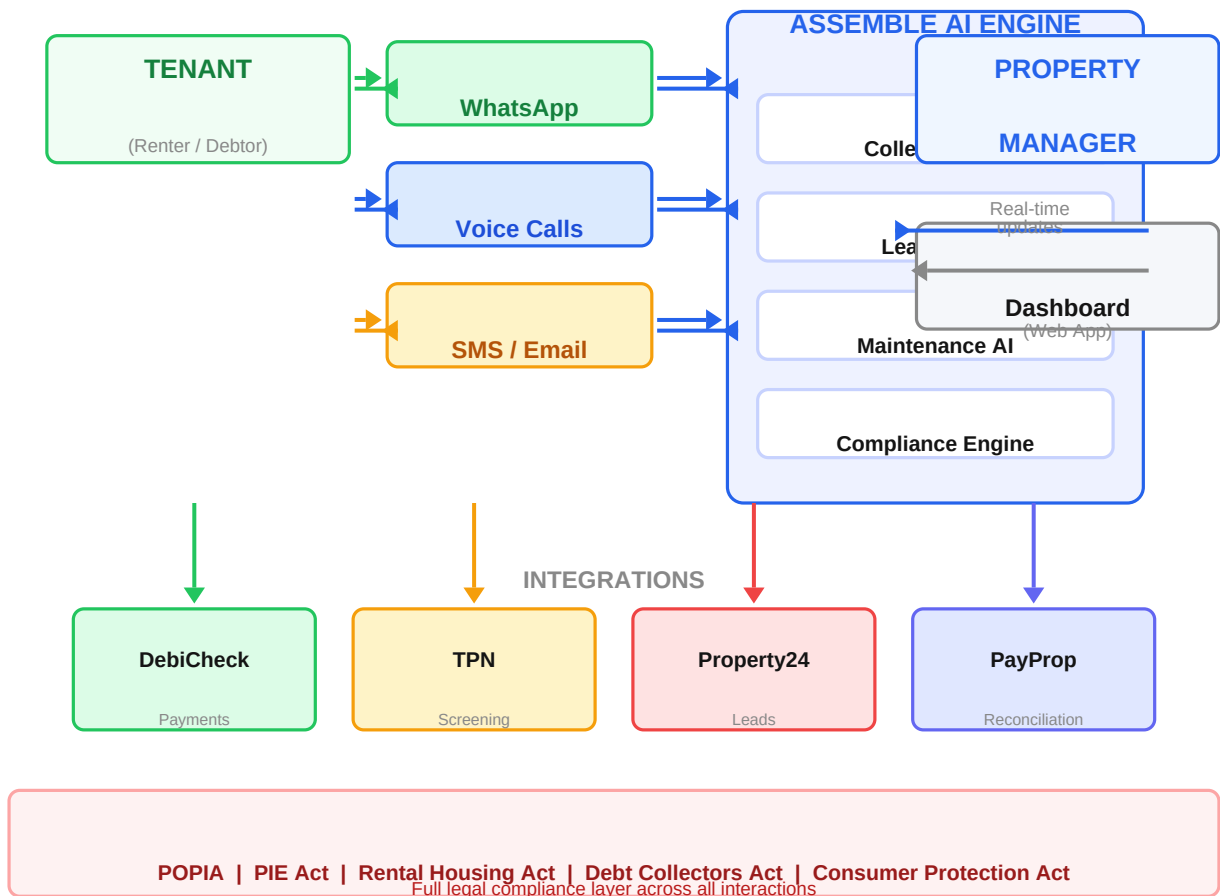
A full property management operations layer that unifies collections, leasing, maintenance, and reporting into a single AI-powered platform:

- Maintenance AI: tenants submit requests via WhatsApp, AI categorises and prioritises, dispatches contractors, tracks resolution
- Tenant portal: payment history, lease documents, maintenance tracking, all accessible via WhatsApp or web
- Owner reporting: revenue dashboards, vacancy tracking, arrears analysis, maintenance cost reporting
- Municipal account verification: automated checking of council bills against official tariffs (unique SA capability)
- AI-generated dashboards: property managers describe what they need in plain English, the platform builds it

How It All Fits Together

The diagram below shows how tenants, communication channels, the AI engine, integrations, and property managers connect within the Assemble AI platform.

ASSEMBLE AI — System Architecture



How It Works

The Collections AI Flow

Property manager uploads tenant/arrears data (CSV, API, or PMS integration)

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Assemble AI segments tenants by arrears depth, payment history, risk

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AI agent initiates contact via WhatsApp (primary channel)

- Personalised message in tenant's preferred language
- Empathetic, professional tone (not threatening)
- Secure payment link included

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Tenant pays
immediately

|

Tenant engages
in conversation

|

No response

|

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Mark resolved
Update PM
dashboard

Negotiate
payment plan
Set up DebiCheck (AI or human)
debit order

Escalate to
voice call

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Still no response

|

Monitor &
auto-collect

Generate letter
of demand
Flag for legal

Compliance Built In

Every aspect of the product is designed around South African legal requirements:

- POPIA: explicit consent management, data minimisation, right to access, appointed Information Officer
- Rental Housing Act (2025 amendments): written lease enforcement, deposit tracking, habitability standards
- PIE Act: automated timeline tracking ensures proper notice periods before any legal escalation
- Debt Collectors Act: if third-party collection is triggered, ensures registration compliance
- Consumer Protection Act: fair terms, proper notice periods, no harassment
- Full audit trail: every message, call, and payment tracked and exportable for legal proceedings

What We Already Have

Assemble AI is not starting from zero. The founding team brings production-grade technology, live customer relationships, and deep domain expertise.

Production Voice AI Platform

We operate a production-grade voice AI platform capable of handling inbound and outbound calls in multiple South African languages. The platform includes telephony integration, speech-to-text, LLM-powered conversation, and text-to-speech. It is live, billing, and handling real calls today for property management clients in Johannesburg.

Municipal Bill Verification Engine

We have built and deployed a verification engine that parses City of Johannesburg municipal bills, extracts line items, and verifies every charge against official tariff schedules with full legal citations. This capability is unique in South Africa and directly relevant to property management operations where municipal accounts are a major cost centre and source of billing errors.

Technical Infrastructure

The existing platform runs on Next.js, Supabase (PostgreSQL), Prisma ORM, and Tailwind CSS, with Anthropic (Claude) and OpenAI integrations for AI capabilities. Authentication, file storage, and deployment infrastructure are all production-ready.

Live Property Management Clients

We have active, paying property management clients in Johannesburg representing tens of thousands of residential units. These relationships provide an immediate pilot base for Collections AI and a direct path to rapid scale.

Active Sales Pipeline

We have contracts in progress with two of the largest property management companies in South Africa, with presentations scheduled across estate agencies, telecoms, insurance, and BPO verticals. The sales infrastructure and relationships are in place.

Market Opportunity

Total Addressable Market

Metric	Value
Renting households in SA	4.5 million
Annual rental flows	R480 billion+
Property management services market	R3.6 billion (growing 9.4% CAGR)
Average national rent	R9,218/month
Average PM fee	10% of monthly rent
National vacancy rate	4.4–5.5% (historic lows)

The Collections Opportunity

Metric	Value
Tenants in arrears	~390,000 households (17%)
Outstanding unpaid rent	R2.7 billion+
Avg cost per non-paying tenant	R28,000–R56,000 (lost rent + legal)
DebiCheck success rate	90–95% (vs 60–70% manual EFT)
AI collections tools in SA	None

Competitive Landscape

The South African PropTech market is fragmented across point solutions:

Player	What They Do	What They Don't Do
PayProp	Rent collection & reconciliation	AI conversations, collections automation
TPN	Tenant screening & credit checks	Collections, leasing automation
WeConnectU	PM software & compliance	AI agents, voice, WhatsApp automation
Prop Data	Agency websites & marketing	Operations, collections, tenant comms
Preferential	AI tenant screening	Collections, leasing, maintenance

Player	What They Do	What They Don't Do
Assemble AI	AI collections + leasing + ops	This is the whitespace we fill

No South African player offers AI-powered conversational collections or leasing. The global leader, EliseAI (\$2.2B valuation, \$100M+ ARR), does not operate in Africa. We are building the EliseAI for this continent.

Revenue Model

Per-Unit SaaS Pricing

Assemble AI charges property managers a monthly fee per unit under management. This aligns our revenue with their portfolio size and scales naturally as they grow.

Product	Price/Unit/Month	Avg PM (500 units)	Large PM (10,000 units)
Collections AI	R15–R30	R7,500–R15,000	R150,000–R300,000
Leasing AI	R10–R20	R5,000–R10,000	R100,000–R200,000
Full Platform	R30–R50	R15,000–R25,000	R300,000–R500,000

Alternative: Performance-Based Pricing

For collections specifically, we can offer a performance-based model where Assemble AI takes 10–15% of recovered arrears. This is zero-risk for the property manager and aligns incentives perfectly. On R2.7 billion in outstanding arrears with a 50% recovery rate, this represents a R168 million annual revenue opportunity at 12.5% commission.

Revenue Projections

Milestone	Units	Avg Price	MRR	ARR
Month 3: First pilots	10,000	R20	R200K	R2.4M
Month 6: Early traction	30,000	R20	R600K	R7.2M
Month 12: Market leader	80,000	R25	R2M	R24M
Month 24: Scale	200,000	R30	R6M	R72M
Month 36: Dominance	450,000	R30	R13.5M	R162M

These projections are conservative. EliseAI reached \$100M+ ARR serving 10% of the US apartment market. South Africa's formal rental market is smaller but the per-unit economics are strong, and Africa expansion (Nigeria, Kenya, Ghana) opens a significantly larger addressable market.

How We Build It

Phase 1: Collections AI (Months 1–3)

Objective: Prove that AI collections works in the SA market

- Build WhatsApp collections agent using existing voice AI platform and WhatsApp Business API
- Integrate with DebiCheck for authenticated payment collection
- Build property manager dashboard (arrears overview, contact history, recovery tracking)
- Deploy with existing property management clients as pilot
- Target: 25–50% reduction in arrears within 30 days of deployment
- Target: 10,000 units under management by end of Month 3

Phase 2: Leasing AI + Scale (Months 4–9)

Objective: Add leasing automation, expand customer base

- Build Property24 and Private Property lead integration (instant response via WhatsApp)
- Add tenant screening workflow (TPN integration, income verification)
- Digital lease generation and e-signature
- Expand to top 20 property management companies nationally
- Target: 50,000 units under management

Phase 3: Full Platform (Months 10–18)

Objective: Become the operating system for SA property management

- Maintenance AI (WhatsApp-based request submission, contractor dispatch, tracking)
- Tenant portal (payment history, documents, communication hub)
- Owner reporting and analytics
- Municipal account verification (leveraging existing Munipal engine)
- AI-generated custom dashboards (property managers describe needs in plain English)
- Target: 150,000+ units, R3M+ MRR

Phase 4: Africa Expansion (Months 18–36)

Objective: Replicate the model across Africa's largest rental markets

- Nigeria (Lagos: 21M people, massive rental market, similar collections challenges)
- Kenya (Nairobi: fast-growing PropTech ecosystem, M-Pesa payment integration)
- Ghana, Tanzania, Rwanda as follow-on markets
- Target: 450,000+ units across the continent

Why We Win

1. WhatsApp + Voice + AI in One Platform

No competitor in South Africa combines conversational AI across WhatsApp, voice, SMS, and email. We do not need to build the voice layer from scratch — it is live and billing today. Adding WhatsApp as the primary collections channel creates a multi-channel system that no point solution can match.

2. Built for South Africa

Multilingual AI (English, Zulu, Sotho, Afrikaans). POPIA-compliant by design. PIE Act timeline tracking built in. DebiCheck integration for reliable collections. Load shedding resilient (WhatsApp is async). Municipal bill verification included. This is not a US product adapted for SA — it is built from the ground up for this market.

3. Distribution Advantage

We are already inside the building. We have paying property management clients and active contracts with the largest property management companies in South Africa. Collections AI is a natural upsell to the same buyer, in the same budget line, with a clear and measurable ROI.

4. Data Flywheel

Every tenant interaction improves the model. We learn which message templates, timing patterns, escalation cadences, and payment plan structures work best for South African tenants. This data compounds over time and creates an advantage that new entrants cannot replicate without years of operational history.

5. Regulatory Moat

South Africa's regulatory environment (PIE Act, Rental Housing Act, POPIA, Debt Collectors Act, Consumer Protection Act) is complex and specific. A US or European product cannot simply be dropped into this market. Our compliance-first approach is a barrier to entry for international players and a differentiator against local competitors who lack the technical capability.

6. Timing

The 2025 Rental Housing Act amendments have just taken effect, creating new compliance requirements that property managers need help with. AI capabilities have reached the point where conversational agents can handle nuanced, multilingual tenant interactions reliably. WhatsApp Business API is mature and widely adopted. The infrastructure moment and the regulatory moment have converged.

What We Need

Assemble AI has the technology, the domain expertise, and the customer relationships to execute. To move from current state to market-leading position, we need:

Technical Build (Months 1–3)

- WhatsApp Business API integration and conversational AI agent for collections
- DebiCheck / payment gateway integration
- Property manager dashboard and reporting
- PMS data integration (CSV import, API connectors)

Go-To-Market (Months 1–6)

- Pilot deployment with existing property management clients (tens of thousands of units)
- Case study development with real arrears recovery data
- Outbound sales to top 20 property management companies in South Africa
- Strategic partnerships with PayProp (payments), TPN (screening), and Property24 (leads)

Team

- Product engineer: full-stack developer with AI/ML experience
- Sales lead: PropTech or SaaS sales experience in SA market
- Compliance advisor: SA property law and data protection expertise

The opportunity is R2.7 billion in unpaid rent.

The technology exists today.

The customers are already ours.

[Let's build the EliseAI for Africa.](#)